

Seat No:

Enrollment No:

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech/Int. B.Tech, Winter 2025-26 Examination

Semester: V / VI / X
 Subject Code: 303105311
 Subject Name: Quant and Reasoning

Date: 26-11-2025
 Time: 2.00 pm to 4.30 pm
 Total Marks: 60

Instructions:

1. This question paper comprises of two sections. Write answer of both the sections in separate answer books.
2. From Section A, **Q.1** is compulsory, From Section B, **Q.1** is compulsory.
3. Figures to the right indicate full marks
4. Draw neat and clean drawings & Make suitable assumptions wherever necessary.
5. Start new question on new page.
6. BT- Blooms Taxonomy Levels – Remember-1, Understand -2, Apply-3, Analyse-4, Evaluate-5, Create-6

SECTION - A

Q.1	Answer the following questions.	Marks	CO	BT
	A. Define the following with one example. (i) Irrational Numbers (ii) Prime Numbers (iii) Composite numbers	06	CO3	BT1
	B. Evaluate the following: (i) Two numbers are in the ratio of 4:5 and the sum of these numbers is 27. Find the two numbers. (ii) Three numbers are in the ratio of 3:4:8 and the sum of these numbers is 975. Find the three numbers. (iii) Two numbers are in the ratio of 4:5. If the difference between these numbers is 24, then find the numbers.	06	CO3	BT3
Q.2	A. A card is drawn at random from a well shuffled pack of 52 cards. Find the probability of getting (i) a jack or a queen or a king, (ii) a two of heart or diamond.	04	CO1	BT4
	B. Twenty women can do a work in sixteen days. Sixteen men can complete the same work in fifteen days. What is the ratio between the capacity of a man and a woman?	05	CO1	BT5
	OR			
	B. Ram celebrated his 4 th birthday on 29 feb 2020. when he was born?	05	CO3	BT5
Q.3	A. Two cars start from the opposite places of a main road, 150 km apart. First car runs for 25 km and takes a right turn and then runs 15 km. It then turns left and then runs for another 25 km and then takes the direction back to reach the main road. In the mean time, due to minor break down the other car has run only 35 km along the main road. What would be the distance between two cars at this point?	04	CO1	BT3
	B. A cyclist moving at a speed of 20 km/hr crosses a bridge in 2 minutes. What is the length of the bridge?	05	CO1	BT5
	OR			
	B. Find the common difference of an A.P. whose first term is 100 and the sum of whose first six terms is five times the sum of the next six terms.	05	CO3	BT2

SECTION - B				
Q.1	Answer the following questions.	Marks	CO	BT
	A. (i) Define the term Average in Mathematics. (ii) State the formula of Arithmetic Mean. (iii) Evaluate the Arithmetic Mean of numbers 12,18, 25, 30 and 35	06	CO1	BT1
	B. (i) If ₹1,200 earns ₹72 as simple interest in 2 years, what is the rate of interest? (ii) Find the rate of simple interest for which a sum of money becomes 5 times of itself in 8 years. (iii) Find the principal, which will amount to ₹ 270.40 in 2 years at the rate of 4% per annum compound interest.	06	CO2	BT3
Q.2	A. How many 5-digit telephone numbers can be constructed using the digits 0 to 9, if each number starts with 67 and no digit appears more than once?	04	CO3	BT2
	B. A, B and C jointly thought of engaging themselves in a business venture. It was agreed that A would invest Rs. 6500 for 6 months, B, Rs. 8400 for 5 months and C, Rs. 10,000 for 3 months. A wants to be the working member for which, he was to receive 5% of the profits. The profit earned was Rs. 7400. Calculate the share of B in the profit.	05	CO3	BT3
	OR			
	B. Read the following information carefully & answer the question below it. A dice is prepared in following manner i. 1 should lie between 2 & 3 ii. 2 should lie opposite to 3 iii. 4 should lie between 5 & 6 iv. 5 & 6 should lie opposite to each other v. 4 should lie face down Que – 1. The face opposite to 1 is? Que – 2. The upper face is?	05	CO3	BT4
Q.3	A. (1) Solve this Alphabet Based Analogy Reasoning AB : L :: BC : ? (2) Find the odd man out 8, 27, 64, 100, 125, 216, 343	04	CO3	BT4
	B. If A is the brother of B, B is the brother of C, and C is the father of D, how is A related to D?	05	CO1	BT4
	OR			
	B. The sum of a non-zero number and twice its reciprocal is $3\frac{3}{4}$. Find the number.	05	CO2	BT3